

6. If we placed the diodes of circuit 3a upside down then what would be the output waveform and why? Draw it.

EXERCISE ^{OR}
6 R – C CIRCUITS
SMOOTHING FILTERS

THEORETICAL PART

R – C circuits are connections of a capacitor C and a resistor R in series, parallel or mixed. An RC circuit represents time. This time is the charging – discharging time of the capacitor and is mathematically proven as follows:

$$C = \frac{Q}{V} = \frac{It}{IR} = \frac{t}{R} \quad t = RC$$

where: C = electrical capacitance
Q = electric charge
V = electrical voltage
R = electrical resistance t
= time

These circuits are mainly used in oscillators to produce waveforms with some, or some, frequency range ($f \propto 1/T$) and in filters. One category of filters is the low-frequency filters and mainly of the frequency of the alternating voltage network of the Public Power Corporation. These filters are part of all power supplies that convert the alternating voltage of the network to direct.

The block diagram of a DC power supply is shown below in Figure 1:

FIGURE 1

We notice that the smoothing filter always comes in after the rectification. Smoothing means balancing, or better, reducing the difference between the maximum and minimum values of the rectifier output voltage. The function of the filter

smoothing is better seen, in figure 2b, if we analyze the circuit of figure 2a.

The circuit analysis is done by examining the circuit in parallel with the diagram that shows us the behavior of the output as a function of the input, in time correspondence. During the positive half-cycle and from time 0 – $T/4$ the diode conducts because it is

correctly polarized and charges the capacitor C (which is at the output) with a value of V_p . From $T/4$ to T the input potential is less than the capacitor potential because the capacitor has not lost its charges as quickly as the input voltage drops and thus the diode is reverse biased and does not conduct. Then the capacitor "sees" only the resistor as a discharge circuit and discharges its charges there. The greater the capacitance of the capacitor or the greater the value of the resistor, the longer the discharge time of the capacitor and therefore the smaller the V_k (ripple voltage), from maximum to minimum output voltage value.

Perhaps the smoothing is better seen if we look at the circuit separately. If it were only the resistor the output would be like simple rectification (waveform V_R in the diagram). If it were only the capacitor then the output would be V_p continuously (waveform V_C in the diagram) because from $T/4$ the capacitor would be charged it would remain charged, since it has nowhere to discharge its charges (with the diode reverse biased from $T/4$ onwards, the capacitor V_p and the input takes values from $-V_p$ to V_p).

We finally observe that the diode constitutes the rectifier stage in the circuit since it converts the input voltage (alternating) into a simply rectified one while the capacitor is essentially a filter. The resistor R does not really exist as a material but represents the input resistance of the next stage of the power supply (voltage stabilizer). So again there is a resistor directly controlled by us. We said that the larger the R or the larger the C , the longer the discharge time, therefore the smaller the ripple V_k . Since R is not directly controlled, then to achieve a small V_k I will have to increase C a lot. And this is how it is done since the indicative values of C in a smoothing filter of this type are 1000 μ f, 4700 μ f, etc.

Simple rectifier with capacitor filter:

The diode conducts: If we consider the diode voltage negligible, then: $V_V t$, 0 C $\ddot{y}\ddot{y}$

The diode in cutoff: The capacitor C is discharged through the resistor R and thus V_0 decays exponentially with a time constant RC . The discharge begins almost at the top of the

V_i . At the end of the interval that lasts as long as the period T , $V_0 V_p V_k$ where V_k is the value of the peak-to-peak ripple.

applies: $V_p V e^{-\frac{T}{RC}}$ Also p

Since $CR \gg T$ we can use the approximation: $e^{-\frac{T}{RC}} \approx 1 - \frac{T}{RC}$

so $V_p V_p V_p (1 - \frac{T}{RC}) V_k V_p \frac{T}{RC}$

The ripple voltage V_k can be expressed as a function of frequency

$f \frac{1}{T}$ as follows:

$V_k = \frac{V_p}{fRC}$	$V_k = \frac{V_p}{2fRC}$
simple correction	Complete restoration

The ripple frequency at full rectification is twice the input frequency.

Input waveform

The average value of the voltage is given by the formula:

$$V_{\text{avg}} = \frac{V_{\text{max}}}{2}$$

Output waveform

The circuit in Figure 3a has a different operating principle and is therefore not used much in power supply smoothing circuits but also in circuits where we simply want to produce a well-smoothed DC component. To better understand the operation, we will first examine what we have at the point where we measure the voltage e .

During the positive half-cycle, diode D1 does not conduct and there is no change in the circuit due to the large value of R1 and R2. During the negative half-cycle, D1 conducts and charges the capacitor to V_{avg} with the polarity

I consider this capacitor as a constant 4 V source. In the next positive alternation I will consider a varying source E at the input as shown in figure 4. I notice that I have two sources in series so at each time point at points A, B I have their sum. In the next negative alternation I will calculate the source E as shown in figure 5.

Source E will be subtracted from 4 V at any time since the sources are in series.

The result is that I have shifted the input voltage towards the positive by 4 V.

FIGURE 4

FIGURE 5

LABORATORY PART

1. Make the circuit of Fig. 7 in your laboratory .

Note: The oscilloscope must be in the DC position on both channels.

- a) Give the input from the test bench: , 8Vpp .

Draw each time a 1KHz sine wave the output waveform of the circuit in time correspondence with the input and measure the average value. What do you observe and why?

- b) Give the input from the test bench a 1 KHz triangular signal , 6 Vpp each time draw the output waveform of the circuit in time correspondence with the input. What do you observe and why?

- c) Reverse the diode of the circuit of figure 7 and repeat steps a and b.

2. a) Make the circuit of figure 2a. Supply the input from 6 Vpp . Draw and observe the output test bench with 1 KHz sine wave signal , waveform in time correspondence with the input. Vary the input frequency. What do you observe? Draw all the waveforms in time correspondence with the input.

- b) Change the resistor value to 10 K Ω and 100 K Ω . Plot the output waveform in time correspondence with the input waveform. What do you observe?

c) Double the capacitor by placing one of its own ($0.1\text{ }\mu\text{F}$) in parallel. Draw the output waveform in time correspondence with the input waveform. What do you observe?

3. a) Implement the circuit of figure 3a, comment and observe the output waveforms (at point C of the circuit) and e (i.e. at point A of the circuit). b) Reduce the value of R_1 to $1\text{ k}\Omega$. What do you observe? with $R_1 = 100\text{ k}\Omega$ and $R_2 = 1\text{ k}\Omega$ what do you observe? Draw all the waveforms in time correspondence with the input

QUESTIONS

1. In the circuit of figure 2a, what changes do fluctuations in frequency bring about? entry and why?
2. What does any increase or decrease in the capacitor cause and why?
3. How will the output waveform of the circuit in Figure 3a change if reverse the diode and why?
4. How do R_1 and R_2 of the circuit in Figure 3a affect the output and why?
5. Consider a simple rectifier with a capacitor filter (see figure 6) which has a sinusoidal input voltage of 100 Hz and a peak voltage $V_p = 4\text{ V}$. The load $R = 10\text{ k}\Omega$. Find the value of the capacitance C that will create a ripple of 0.2 V peak to peak.

EXERCISE ^{OR}
7 SCISSORS

THEORETICAL PART

Clippers are the circuits that we use to cut off the peaks of a waveform at a specified level, or allow part of it to be carried. This is easily achieved if we place a DC voltage source in series with the clipper diode and a suitable one, so that the diode becomes conductive above this voltage. Let's look at the circuit of Figure 1A, a positive voltage clipper.

It is clear that up to the value of 2.6V (t_0 to t_1) of the alternating voltage applied to the input of the circuit, the diode is reverse biased since the anode is more negative than the cathode and no current flows through the diode. The output is then equal to the input (t_0 to t_1)

When the input voltage becomes equal to 2.6V the diode becomes conductive (there is a potential difference of 0.6V between the anode and cathode), and at the output we have the source voltage $E=2V$ constant voltage + 0.6V of the diode. This will happen until time t_2 when the input voltage drops to 2.5V and the diode is cut off. The output again follows the input.

Transfer characteristic

The curve that relates the output voltage U_o of a circuit to the input U_i is called the transfer characteristic. The transfer characteristic of the above circuit is shown in Figure 2.

Figure 2

LABORATORY PART

1. Implement the circuit of Figure 1b in your test bench. Apply an alternating voltage of 8 V_{pp} with a frequency of 1 KHz and observe the output waveform.
2. Vary the DC voltage of source E. What do you observe?
3. Connect the oscilloscope so that you can see the transfer characteristic on the screen. Explain its shape.
4. Let us now reverse the diode. The circuit of Figure 3 a, b results.

5. Apply an alternating voltage of $8 V_p$ to the input . Draw the output waveforms, as well as the transfer characteristic. Explain in a few words the operation of the circuit. Repeat the above for the circuits of figures 4 (a or b), 5, 6 in paragraphs 5a) , 5b) and 5c)

QUESTIONS

1)

Shredder circuit